

## Exercises

Consider a database, which contains the details of human resource for a company with four relations Location, Job details, Department and Employee given in pages 4, 5, and 6. The relations show some values for human resource database of a company. The relations are uniquely identified by the attributes Location Id, Job Id, Department Id and Employee Id respectively. Each of the departments are located in different places and employees are serving in various job roles in different departments.

**Location** (Location\_ID, City)

**Department** (Department\_Id, Department\_name, Manager\_id, Location\_Id)

**Job\_Details** (Job\_Id, Job\_Title, Min\_Salary, Max\_Salary)

**Employee** (Employee\_Id, First\_name, Last\_name, Email, Hire\_date, Salary, Job\_Id, Department\_Id)

Create a database Human\_Resource with the above mentioned four relations by clearly specifying the corresponding primary key and foreign key constraints in addition to the conditions described in the table below for each of the relations.

| Relations   | Conditions                                                                                                                                                                                                           |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location    | All of the attributes should not be null.<br>Set the default value for the <b>city</b> attribute as Vavuniya                                                                                                         |
| Job_Details | The attributes Job_ID and Job_title should not be null and Max_salary and Min_salary must be greater than zero.<br>Set the default value for the minimum salary as LKR 5,000 and the maximum salary as LKR 100, 000. |
| Departments | All of the attributes should not be null.<br>Set the default department name as HR                                                                                                                                   |
| Employee    | The attributes First_name, Last_name, and Email must not be null and salary should be greater than zero.                                                                                                             |

Answer the following questions with suitable MySQL statements:

- Replace all the null values on the Hire\_date field of the employee table with 2008-03-17 using IFNULL function.

- ii. Check whether anyone has the same First\_name and Last\_name in the employee relation.
- iii. Check whether any job has the same max\_salary and min\_salary values in the Job\_Details relation.
- iv. Insert a record into the Job\_Details table by passing the default values for minimum and maximum salary.

Write MySQL statements to prepare the index for each of the following:

- i. Create index for the employee relation on the attributes First\_name and Last\_name.
- ii. Create index on the attributes Department\_Id for the Employee relation.
- iii. Create a composite index on the attributes Job\_title and Max\_salary for the Job\_Details relation.
- iv. Create a composite index on the attributes First\_name and Email for the Employee relation.
- v. List down the relational\_schema, relation\_name, index\_name, and column\_name of created indexes for the database Human\_Resource.